DEEPAK KATUKURI

SOFTWARE ENGINEER · FULL-STACK · AI-CURIOUS



CONTACT

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TECHNICAL SKILLS

Languages:
Python, TypeScript / JavaScript, SQL, PL/SQL, Core Java, C++

Full-Stack:
Next.js, React, TypeScript, Vite, Tailwind CSS, React Query, Framer Motion, PostgreSQL, FastAPI, Recharts

LLM & RAG:
Retrieval-Augmented Generation, LangChain, hybrid retrieval (BM25 + dense, RRF), reranking, pgvector, prompt engineering, Ollama

Deep Learning:
PyTorch, TensorFlow, CNNs, EfficientNet, fine-tuning, Hugging Face

Machine Learning:
Scikit-Learn, classification, ensembles, NumPy, Pandas, Matplotlib, Seaborn

MLOps & Cloud:
Docker, Kubernetes, AWS, CI/CD, model deployment

Tools:
Git/GitHub, code review, pytest, Jupyter, IBM Watson

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

LANGUAGES

English – Fluent

German – A1/A2 (learning)

PROFILE

Software Engineer with nearly 3 years of production experience and a full-stack, TypeScript/React skill set (Next.js, FastAPI, PostgreSQL), currently deepening AI/ML depth via an MSc in Data Science. Comfortable evaluating and adopting new dev tooling on evidence: hand-rolled a hybrid RAG retriever, then rebuilt it on LangChain to test the framework against a fixed benchmark, catching and fixing a silent regression before trusting it at scale. Builds real products end to end — from a full-stack health app (React + FastAPI + PostgreSQL) to a Next.js/TypeScript portfolio site — and is curious about applying AI-native development practices to real engineering work. Seeking an English-speaking working-student Software Engineering role in Germany.

PROJECTS

RA Nutrition & Lifestyle App – Full-Stack Health Automation

React · FastAPI · PostgreSQL · Recharts · personal product build

- Translated operational health-tracking needs into a shipped full-stack feature: React/Vite/Tailwind frontend over a FastAPI + PostgreSQL backend (4 logical schemas), with JWT auth and token refresh.
- Built a rule-based recommendation engine and a weekly analytics dashboard (Recharts) that closes the loop from raw meal/symptom logs to actionable output for the end user.
- Architected (not yet shipped) an ML forecasting tier — scikit-learn/XGBoost and PyTorch LSTM/GRU/TCN — for flare-risk prediction, scoping the next iteration before building it.

Developer Portfolio – Next.js & TypeScript

Next.js 15 · TypeScript · React 19 · Tailwind CSS · Framer Motion · Radix UI · github.com/Deepak589/NextJs-Portfolio

- Built and shipped a personal portfolio site on the Next.js App Router with strict typing (`tsc --noEmit` in the build script), architecting reusable UI primitives (Button, Card, Badge) on Radix UI.
- Implemented custom interactive components – a radial orbital timeline, tilt cards, animated typography – using Framer Motion and Tailwind CSS v4.

RAG Pipeline – Hand-Rolled Retrieval, Validated Against LangChain at Scale

Python · LangChain · BM25 + dense hybrid · pgvector · Ollama · github.com/Deepak589/RAG_pipeline

- Built a hand-rolled hybrid retriever (BM25 + dense bi-encoder, RRF fusion, cross-encoder reranking) from scratch across 4 iterative stages, reaching 0.833 recall@5 / 0.672 MRR on a 55-question labeled golden set before reaching for a framework.
- Rebuilt the same retriever on LangChain over the identical corpus and labels, changed only the framework, and caught a silent 0.06-recall regression from LangChain's default BM25 tokenizer – fixed it to 0.854 recall@5, confirming framework parity before trusting it at scale.
- Scaled the corpus ~100x (7 to ~740 documents, ~3.5GB across arXiv, SEC filings, Wikipedia, and scanned PDFs) with a resumable, file-type-routed ingestion pipeline feeding a pgvector-backed dense store.

EXPERIENCE

Software Engineer — Mar 2022 – Nov 2024

ValueMomentum

- Maintained 2 enterprise P&C insurance and surety-bond applications on OneShield PAS, resolving production defects across policy workflows.
- Migrated full policy, claims, premium, and customer history for the Surety Bonds line from a legacy policy admin system into OneShield PAS, building and validating PL/SQL migration scripts across customer, policy (PID), claims, and premium tables.
- Root-caused a recurring data-integrity defect where claims data for pre-2019 Surety bonds under one bond type loaded empty – the OneShield object copy from the stored object to the loading object silently failed when another bond shared the same bond type under a different bond ID – and kept records current with a weekly backfill script pending a permanent fix.
- Translated business requirements from underwriting and claims stakeholders into technical specs during sprint planning and code reviews, then accelerated releases by running QA/UAT test cases and supporting CI/CD deployment across the full policy lifecycle.

ML Intern — Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it via IBM Watson ML, communicating results to non-technical reviewers with Matplotlib/Seaborn visualizations.